

Electricity Demand Forecasting for Bangladesh Using Machine Learning and Deep Learning

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Abstract—Power-system planning, generation scheduling, resource allocation, and dependable grid operation all depend on accurate electricity-demand forecasts. Using hourly electricity-demand data from the Power Grid Company of Bangladesh, this study develops a data-driven forecasting system for Bangladesh. Weather variables are not used in this experiment; instead, the system uses time-based, peak-period, seasonal, and lagged-demand features extracted from the demand series. Thirteen input features and 94,766 observations make up the final dataset. Five models are assessed: Multi-Layer Perceptron (MLP), Random Forest, XGBoost, Linear Regression, and Simple Recurrent Neural Network (RNN). The training, validation, and independent test phases of the data are separated chronologically.

MLP provides an RMSE of 1692.42 MW and an R^2 of 0.6307 on the test set, whereas RNN provides the lowest RMSE of 1665.20 MW and the highest R^2 of 0.6425.

Index Terms—electricity demand forecasting, Bangladesh, machine learning, deep learning, MLP, RNN, time-series forecasting

I. INTRODUCTION

Electricity-demand forecasting plays an important role in power-system planning and operation. Accurate forecasts help electricity providers maintain a balance between supply and demand, schedule electricity generation, allocate resources, and reduce operational costs. Forecasting errors may result in either insufficient electricity generation or unnecessary resource allocation. Electricity consumption is influenced by historical demand, time-related patterns, weather conditions, population growth, and economic activity. In Bangladesh, high temperatures and humidity can increase the use of cooling appliances, while rainfall and wind conditions may also affect daily activities and electricity-consumption behavior. Consequently, historical demand alone may not provide sufficient information for an accurate weather-sensitive forecasting system. The objective is to evaluate deep learning and traditional machine learning models for estimating Bangladesh's hourly power consumption. Thirteen input features—hour, month, year, peak-period indicators, day-type indicators, season indicators, and two lagged-demand variables—are used in the present code, which is derived from the demand dataset. Because supply-side variables may include information that is closely connected to the aim and might cause leakage, they are eliminated. The research employs a rigorous time division: training from 2015 to 2020, validation from 2021 to 2022, and

testing starting in 2023. Finding the best model for the later unobserved era and figuring out what can be inferred from the comparison are the primary goals.

II. LITERATURE REVIEW

Neural networks, machine learning models, and statistical techniques have all been used to study electricity-demand predictions. The following studies are pertinent because they offer objectives, model selections, performance outcomes, gaps, and practical insights for the ongoing demand-only project.

A. Weather and Electricity Demand in Dhaka

Istiaque and Khan used approximately 16,508 observations collected from 2011 to 2017 to investigate the relationship between Dhaka's ambient temperature and electricity consumption. Their objective was to understand how changes in temperature influence electricity demand. They applied several statistical techniques, including curve fitting, regression analysis, covariance analysis, Pearson correlation, and time-series forecasting. The study reported a Pearson correlation coefficient of approximately 0.758 between temperature and electricity consumption. The authors also estimated that a 1°C decrease in temperature could reduce electricity demand by approximately 81 MW. However, although electricity demand is influenced by several factors, the final forecasting relationship mainly relied on temperature. The study supports the idea that electricity demand is affected by significant time-dependent and external factors. However, weather-related variables are not included in our project. Therefore, our existing model evaluates how effectively historical electricity demand and time-based features alone can provide meaningful forecasting insights.

B. LSTM-Based Forecasting for Chattogram

An LSTM model was proposed for short-term electricity-load forecasting in Chattogram [2]. The goal was day-ahead hourly forecasting using national PGCB demand and temperature data. Chattogram's electricity demand was estimated as 11% of the national demand. The LSTM model was compared with a Support Vector Machine and achieved an MAE of 92.448 MW and an RMSE of 127.682 MW. The main limitations included estimated regional demand, limited

weather information, manual model selection, and the absence of a separate validation period. Our project directly uses national demand data and compares five models with separate validation and test periods. The main learning is that sequence models can be useful, but realistic temporal testing is also important.

C. CNN and Stacked BiLSTM Forecasting for Dhaka

For short-term electricity-demand forecasting in Dhaka, Akhter et al. developed a hybrid CNN and stacked BiLSTM model [3]. Their aim was to combine local pattern extraction with sequential learning. Using 2,190 daily observations, they compared the CNN-BiLSTM model with LSTM and CNN-LSTM variants. The proposed approach achieved an MSE of 0.0151, an RMSE of 0.1229, an MAE of 0.0929, and a MAPE of 1.6445%. The primary limitations were the use of daily rather than hourly data and the absence of directly integrated weather, holiday, demographic, and socioeconomic variables. Our project uses hourly national demand data and simpler, more reproducible models. The main lesson is that complex models should be compared with simpler baselines on the same forecasting problem.

D. Improved Forecasting of National Demand

Using environmental, economic, historical, and temporal data, Islam, Turja, and Habib proposed an improved demand-forecasting framework for Bangladesh [4]. Their objective was to enhance national forecasts through feature engineering and model comparison. They used 78,906 hourly PGCB records, applied the interquartile range (IQR) method to remove 344 outliers, and evaluated Random Forest, LSTM, and FB-Prophet using multiple scaling techniques. FB-Prophet achieved a validation MAPE of approximately 2.28%, followed by LSTM with Min–Max scaling at 2.33% and Random Forest at 3.03%. A limitation was that the environmental variables consisted primarily of temperature and humidity represented as daily averages for hourly demand. Our study intentionally excludes weather variables and uses temporal, seasonal, peak-period, and lagged-demand information to establish a demand-only baseline. The main lesson is that feature engineering and temporal evaluation can significantly affect forecasting performance.

E. Hybrid Forecasting with Multiple Variables by Saranj and Zolfaghari

They introduced a hybrid electricity-consumption forecasting framework that integrates LSTM, Random Forest, ARIMAX-GARCH, and an Adaptive Wavelet Transform [5]. Their objective was to improve forecasting accuracy by incorporating several external variables. Using 56,927 half-hourly observations from the United Kingdom, the proposed framework achieved RMSE values of 1.536 in summer and 1.249 in winter, along with MAPE values of 1.276% and 0.869%, respectively. However, its complex architecture and application to a different electricity market limit its direct suitability for Bangladesh. In contrast, our project adopts a simpler and

more reproducible methodology. The study demonstrates that historical load and temporal information can support effective forecasting without requiring an excessively complex hybrid system.

F. Comparative Forecasting with XGBoost and Deep Neural Networks

The study compared Prophet, XGBoost, and deep neural networks using 55,576 hourly observations from Türkiye [6]. Its goal was comparative forecasting using temporal, holiday, lagged, and rolling features with chronological evaluation. XGBoost achieved MAE 981.48 MWh, RMSE 1344.15 MWh, and MAPE 2.72%, while the deep neural network achieved MAE 997.82 MWh, RMSE 1306.96 MWh, and MAPE 2.77%. Prophet performed much worse. The study also emphasized that random cross-validation can leak future information. Our code follows the same important principle by using chronological train, validation, and test periods. The learning is that validation strategy is important when comparing forecasting models.

G. MLP-Based Electricity Load Forecasting

Using actual power-system data from four European nations, Dudek investigated MLP-based short-term load forecasting [7]. The objective was to investigate the effects of data decomposition and representation on MLP accuracy. The study contrasted MLP with self-organizing-map models, fuzzy counter-propagation networks, radial-basis-function neural networks, and generalized regression neural networks. Forecast error was greatly decreased by using more local MLP techniques. For the most local form, the best MAPE values were 1.16% for Poland, 2.57% for Great Britain, 1.62% for France, and 1.27% for Germany. The difference with our project is that our code employs one MLP over Bangladesh demand features, whereas it employed a separate decomposition approach and nations. The key takeaways are that temporal representation can have a significant impact on performance and that MLP is a reliable electricity-load forecasting model.

H. Bangladesh Hourly Electricity Dataset

The “Hourly Electricity Generation, Demand, Load Shedding, and Fuel Mix Dataset for Bangladesh (2015–2026)” was published in 2026 to provide structured hourly data collected from the PGCB public ERP portal [8]. Covering the period from April 2015 to March 2026, it contains more than 92,000 records on electricity demand, generation, load shedding, fuel-based generation, and power imports. For the present study, supply-side variables are excluded to construct a demand-only forecasting dataset. This dataset offers a valuable national-level foundation for the reproducible evaluation and comparison of multiple forecasting models.

I. Research Gap and Motivation

Previous studies consistently highlight the importance of historical demand, temporal patterns, effective feature engineering, and rigorous validation in electricity-demand forecasting. Research conducted in Bangladesh further demonstrates

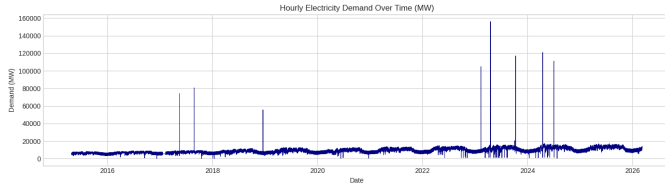


Fig. 1. Hourly electricity demand over the complete dataset.

the relevance of weather conditions; however, weather data are not available in the current implementation. Consequently, this study establishes a demand-only baseline and evaluates Linear Regression, Random Forest, XGBoost, MLP, and RNN using a consistent chronological data split and scaling procedure. This approach determines the forecasting capability of historical demand and calendar-based features before weather variables are incorporated in future research.

III. DATASET AND DATA PREPARATION

This study uses the PGCB-associated hourly electricity dataset for Bangladesh, which spans April 2015 to March 2026 and includes more than 92,000 hourly observations [8]. During preprocessing, the notebook converts the relevant field into datetime format, arranges the records chronologically, and discards unnecessary supply-side variables.

Following the removal of unused fields, the initial working dataset consists of 94,934 observations and two primary variables. The feature-engineering process reduces the dataset to 94,766 observations with 15 columns, comprising electricity demand in MW as the target and 13 predictor variables.

The engineered predictors include hour, month, year, day-peak, evening-peak, night, weekend, weekday, three seasonal indicators, one-hour demand lag, and 168-hour demand lag. Day-peak hours are defined as 09:00–16:59, evening-peak hours as 17:00–22:59, and nighttime hours as 23:00–05:59. The seasonal categories are Winter from December to February, Summer from March to June, and Monsoon from July to November.

The one-hour lag captures demand from the immediately preceding observation, whereas the 168-hour lag approximates demand at the corresponding hour of the previous week. Observations containing missing values introduced by these lag features are discarded. Generation, fuel-specific generation, electricity imports, load shedding, and other supply-side variables are excluded because of their close association with observed demand and the resulting risk of data leakage.

IV. EXPLORATORY DATA ANALYSIS

The first rows, data information, and descriptive statistics are used in the notebook to analyze the cleaned data. Demand ranges from 6 MW to 156,050 MW, with a mean of 9013.16 MW and a standard deviation of 2833.82 MW. The unusually high maximum indicates the presence of extreme observations. The demand series contains several large spikes and substantial fluctuations (Fig. 1).

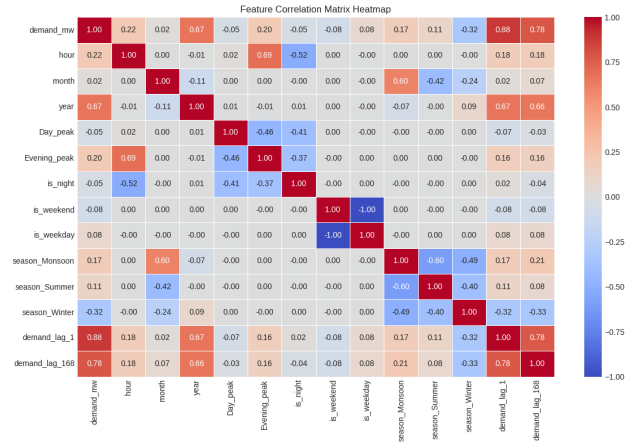


Fig. 2. Pearson correlation matrix of the numerical variables used in the notebook.

Strong positive correlations between demand and past demand are displayed in the correlation heatmap in Fig. 2. The correlation is around 0.88 with demand lag 1 and 0.78 with demand lag 168. The year feature and demand have a correlation of roughly 0.67. Correlation does not establish causation; it is merely a tool for investigation.

V. METHODOLOGY

The complete workflow is shown in Fig. 3. It starts with PGCB hourly demand data and continues through cleaning, feature engineering, EDA, chronological splitting, leakage-safe scaling, model training, and evaluation.

A. Chronological Split and Scaling

There are 49,387 training samples from 2015 to 2020, 17,498 validation samples from 2021 to 2022, and 27,881 testing samples from 2023 onward. Only training inputs are used to fit StandardScaler, which is subsequently applied to validation and test inputs. Only when training is required is a different target scaler installed. Prior to the final assessment, predictions are converted back to MW.

B. Forecasting Models

For the 13 inputs, the baseline model is linear regression. Random Forest employs parallel training, random state 42, and 50 trees. Using validation data, XGBoost employs 200 estimators, a learning rate of 0.1, a random state of 42, and early halting after 10 rounds.

MLP employs dense layers with a single linear output and 64, 32, and 16 hidden units. Hidden layers make advantage of ReLU. Adam, batch size 32, mean squared error, and 20 epochs are employed.

RNN employs thick layers with 16 units and a single output after a SimpleRNN layer with 32 units. Adam, batch size 32, mean squared error, and 20 epochs are employed. Since each row is transformed into a sequence with a single time step, the constructed lag characteristics provide the majority of the temporal information.

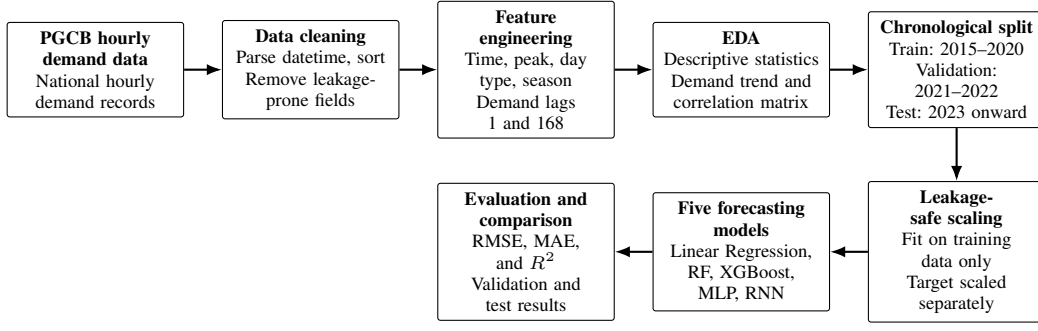


Fig. 3. Complete electricity-demand forecasting workflow used by the current notebook.

C. Evaluation Metrics

Three metrics are used: RMSE, MAE, and R^2 .

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}. \quad (1)$$

Here, n is the number of observations, y_i is the actual demand for observation i , and $\hat{y}_i$ is the predicted demand. Lower RMSE is better.

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|. \quad (2)$$

Lower MAE is better.

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}. \quad (3)$$

Here, $\bar{y}$ is the mean actual demand. Higher R^2 is better.

VI. RESULTS AND DISCUSSION

Table I summarizes the final performance of the evaluated models on the test set. RNN achieves the strongest overall test performance, producing the lowest RMSE of 1665.20 MW and the highest R^2 of 0.6425. MLP follows closely with an RMSE of 1692.42 MW and an R^2 of 0.6307. Although MLP records the lowest validation RMSE of 409.27 MW, RNN generalizes more effectively to the independent test period. This result indicates that superior validation performance does not necessarily ensure the best performance on later unseen data.

Linear Regression obtains a test RMSE of 1977.33 MW and an R^2 of 0.4959. However, its MAE of 495.06 MW is lower than those of MLP, XGBoost, and Random Forest. Among the evaluated approaches, XGBoost and Random Forest demonstrate comparatively weaker performance during the final test period.

A. Validation-to-Test Generalization

Each of the five models exhibits a substantial increase in RMSE between the validation and test periods. The RMSE of RNN rises from 439.83 MW to 1665.20 MW, whereas that of MLP increases from 409.27 MW to 1692.42 MW. These results suggest thdemand during the later test period is considerably more difficult to forecast.

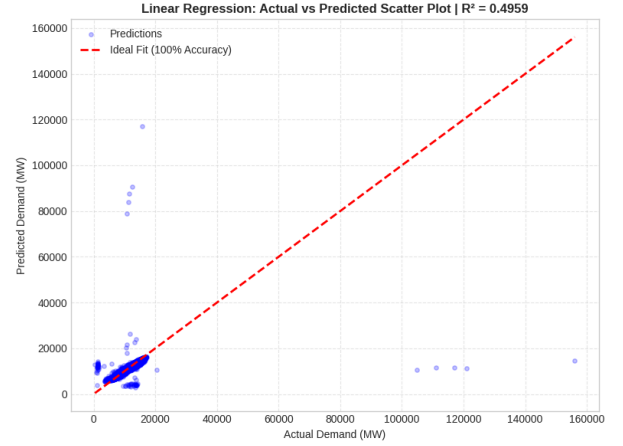


Fig. 4. Linear Regression actual versus predicted demand.

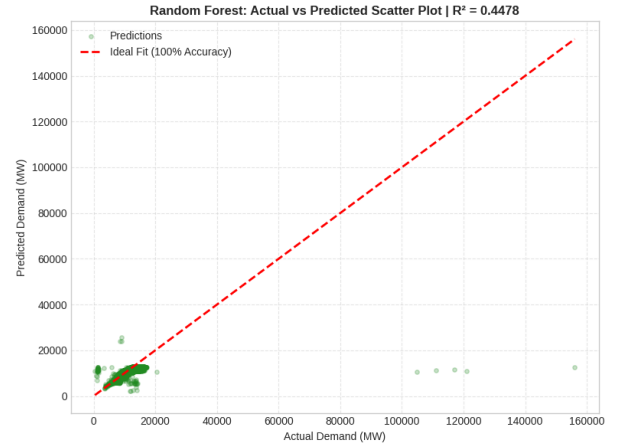


Fig. 5. Random Forest actual versus predicted demand.

The demand series also includes extreme values reaching 156,050 MW. Because the current notebook does not apply an explicit outlier-treatment procedure, such observations may have a considerable influence on RMSE and R^2 . In addition, changes in demand behavior between the training and later test periods may contribute to the observed decline in generalization performance.

TABLE I
FINAL MODEL COMPARISON ON THE TEST SET

Model	Val. RMSE (MW)	Test RMSE (MW)	Test MAE (MW)	Test R^2
RNN	439.83	1665.20	487.87	0.6425
MLP	409.27	1692.42	573.59	0.6307
Linear Regression	523.43	1977.33	495.06	0.4959
XGBoost	530.62	2022.34	951.06	0.4727
Random Forest	610.89	2069.43	1010.75	0.4478

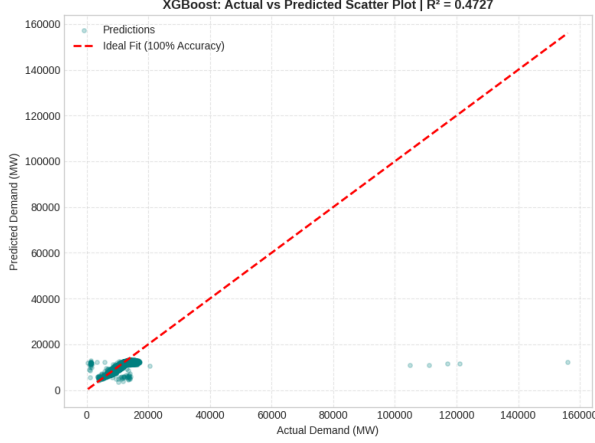


Fig. 6. XGBoost actual versus predicted demand.

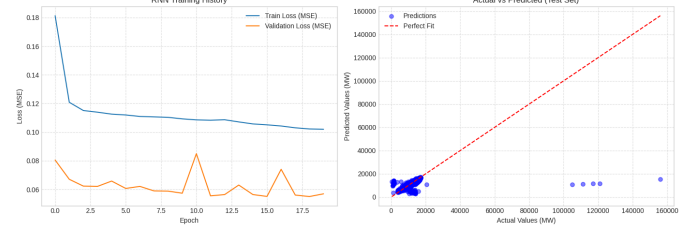


Fig. 8. RNN training history and actual-versus-predicted test demand.

imports TimeSeriesSplit but does not run a time-series cross-validation loop. There is no specific handling of outliers. Sequence length one is used by the RNN. MAPE is not computed in the current notebook, and model settings are primarily adjusted by hand.

VIII. FUTURE WORK

Future research can compare the demand-only baseline with a weather-aware model and incorporate hourly weather variables. RNN or LSTM can be evaluated on longer sequences. Random Forest, XGBoost, and MLP can all be tuned more methodically. It is also possible to include explicit outlier analysis and expanding-window validation.

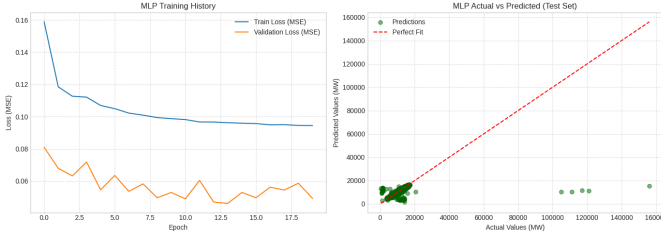


Fig. 7. MLP training history and actual-versus-predicted test demand.

B. Learning from the Current Experiment

Demand lags 1 and 168 are powerful forecasting indicators, according to the correlation analysis. This confirms the findings of the literature that there is a significant time dependence on power demand. Additionally, the comparison demonstrates that when useful latency features are already available, MLP can perform similarly to RNN.

Onetime step is given to each row of the current RNN. As a result, it cannot be referred to as a complete multi-hour sequence model. The engineered features are mostly responsible for its successful outcome.

VII. LIMITATIONS

The added value of temperature, humidity, rainfall, and wind cannot be measured because the current experiment does not account for weather factors. The findings come from a single fixed chronological train-validation-test split; the notebook

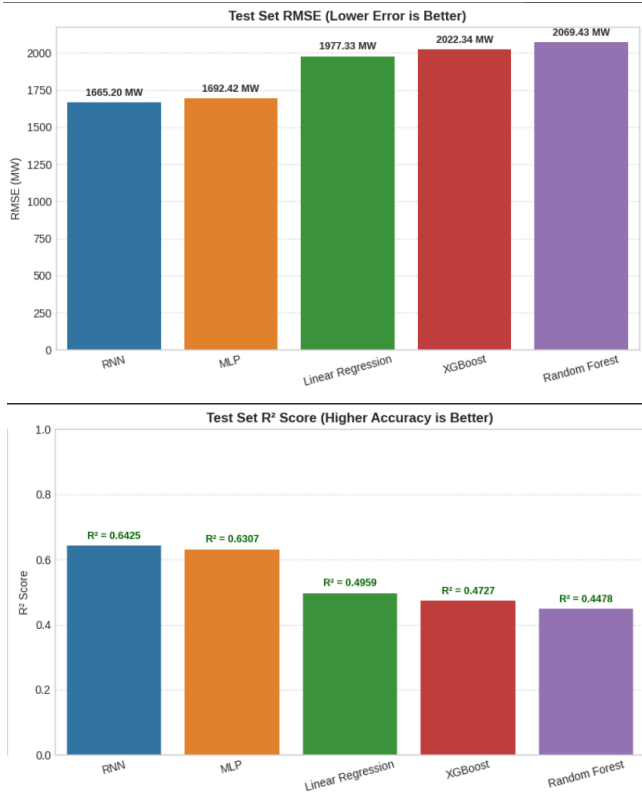


Fig. 9. Test RMSE and test R^2 comparison. Each model keeps a distinct color across both panels, following the notebook.

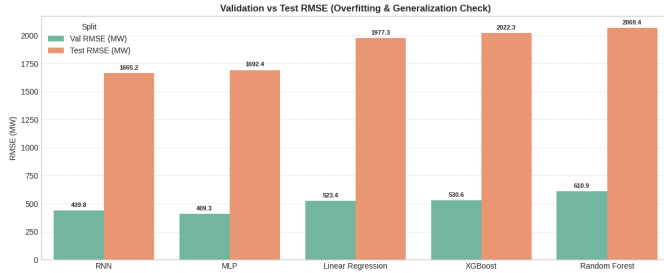


Fig. 10. Validation versus test RMSE. The large increase indicates a generalization gap.

IX. CONCLUSION

This study uses hourly demand data without meteorological variables to propose a straightforward and repeatable electricity-demand forecasting framework for Bangladesh. The existing code compares Linear Regression, Random Forest, XGBoost, MLP, and RNN while generating 13 input features from time, peak-period, day-type, season, and lag data. With RMSE 1665.20 MW and R^2 0.6425, RNN outperforms MLP, which yields RMSE 1692.42 MW and R^2 0.6307. The findings demonstrate that neural models are capable of capturing nonlinear relationships in the constructed feature set and that historical demand features are very helpful. The significant validation-to-test disparity further demonstrates how challenging it is to forecast demand behavior in the future.

Future weather-aware and longer-sequence studies can make advantage of this demand-only baseline.

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